

Dean, Principal & HOD Executive Dashboard API

Complete Technical Integration Guide & Data Contracts for React Web Developers

Document Version: 1.0 (Production)	Target Base URL: http://172.29.58.78:3000
Release Date: September 2026	Auth Protocol: JWT Bearer Token (Header: Authorization)
Target Audience: Frontend React Engineer	Access Roles: dean, principal, hod, admin

1. System Architecture & Authentication

This API enables the GKCE Management Web Portal (accessible by **Dean Ma'am, Principal Sir, and Department HODs**) to monitor real-time class attendance, individual student percentage records, section-by-section progress, faculty attendance submission compliance, and defaulters with attendance < 75%.

Authentication Scheme: All requests to /api/dashboard/* must pass a valid JSON Web Token in the HTTP Authorization header:

Authorization: Bearer <JWT_TOKEN>

Role	Dashboard Scope & Permissions	Permitted Routes
dean	Full college-wide visibility across all 5 departments (CSE, ECE, EEE, MECH, CIVIL), all sections, all students, and executive analytics.	All dashboard & export endpoints
principal	Executive access identical to Dean; can inspect institutional trends, college attendance rates, and download consolidated registers.	All dashboard & export endpoints
hod	Departmental jurisdiction; default queries scoped to assigned department (e.g. CSE). Can view department sections, students, and faculty.	All dashboard endpoints (filtered to dept)
admin	System management, student registration, role assignment, and master audit.	All endpoints

2. Complete Endpoints Quick Reference

Method	Endpoint Route	Primary Purpose	Access
POST	/api/auth/login	Authenticate Dean, Principal, HOD, Teacher	Public
GET	/api/auth/me	Get current authenticated user profile & role	Auth
GET	/api/dashboard/overview	College/Dept KPI counters, attendance rate, submissions	Executive
GET	/api/dashboard/departments	Branch-wise comparison (CSE, ECE, EEE, etc.)	Dean/Princ/Admin
GET	/api/dashboard/sections	Sections list, faculty in charge, today's status	Executive
GET	/api/dashboard/section-details/:class	Single section matrix: students, present/absent roll numbers	Executive
GET	/api/dashboard/students	Student directory with attendance %, search & filters	Executive
GET	/api/dashboard/student/:id	360° student profile: monthly logs, parent phone, absences	Executive
GET	/api/dashboard/defaulters	Low attendance list (< 75%) with shortage calculation	Executive
GET	/api/dashboard/faculty-status	Faculty compliance: submitted vs pending classes today	Executive
GET	/api/dashboard/trends	Time-series analytics for Recharts graphs (7d/30d/month)	Executive
GET	/api/dashboard/export/excel	Download official multi-tier Excel register (.xlsx)	Executive

3. Detailed Endpoint Specifications & Payloads

The following contracts contain exact JSON keys and types designed for seamless state mapping in React.

3.1 Authentication: Login & Profile

POST /api/auth/login

Authenticates faculty or administration. Returns a Bearer JWT valid for 7 days.

```
// Request Body (JSON)
{
  "email": "dean@college.edu",          // or principal@college.edu, hod.cse@college.edu
  "password": "DeanPass123!"
}

// Response (200 OK)
{
  "success": true,
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "user": {
    "id": "66da9123f8b1c4e901a1bc30",
    "name": "Dr. Sarah Dean",
    "email": "dean@college.edu",
    "role": "dean",                      // "dean" | "principal" | "hod" | "admin" | "teacher"
    "department": "ALL"                 // or "CSE", "ECE", etc.
  }
}
```

3.2 Executive KPI Overview

GET /api/dashboard/overview?department=ALL&date;=2026-09-19

Generates top-level cards for Total Enrolled Students, Today's Attendance %, Sections Breakdown, and Pending Submissions.

```
// Response (200 OK)
{
  "success": true,
  "data": {
    "kpi": {
      "totalStudents": 420,
      "totalSections": 12,
      "totalDepartments": 5,
      "todayAttendanceRate": 88.5,      // Overall college percentage today
      "monthlyAverageRate": 84.2,      // Average across the current month
      "todayPresentCount": 372,
      "todayAbsentCount": 48,
      "defaultersCount": 18,           // Students below 75%
      "criticalCount": 4               // Students below 50%
    },
    "submissionCompliance": {
      "totalClasses": 12,
      "submittedCount": 10,
      "pendingCount": 2,
      "complianceRate": 83.3          // 10/12 submitted
    },
    "date": "2026-09-19",
    "department": "ALL"
  }
}
```

3.3 Department-wise Comparative Breakdown

GET /api/dashboard/departments?date=2026-09-19

Provides Dean and Principal with a side-by-side performance table of all branches.

```
// Response (200 OK)
{
  "success": true,
  "departments": [
    {
      "code": "CSE",
      "name": "Computer Science & Engineering",
      "hodName": "Dr. K. Ramesh",
      "hodEmail": "hod.cse@college.edu",
      "totalStudents": 180,
      "totalSections": 4,
      "todayAttendanceRate": 92.4,
      "monthlyAverageRate": 89.1,
      "submittedSections": 4,
      "pendingSections": 0,
      "defaultersCount": 5
    },
    {
      "code": "ECE",
      "name": "Electronics & Communication Engineering",
      "hodName": "Dr. V. Lakshmi",
      "hodEmail": "hod.ece@college.edu",
      "totalStudents": 120,
      "totalSections": 3,
      "todayAttendanceRate": 86.0,
      "monthlyAverageRate": 82.5,
      "submittedSections": 2,
      "pendingSections": 1,
      "defaultersCount": 8
    }
  ]
}
```

3.4 Sections List & Submission Status

GET /api/dashboard/sections?department=CSE&year;=1&date;=2026-09-19

Lists all class sections with assigned faculty in-charge, today's submission status, and attendance percentages.

```
// Response (200 OK)
{
  "success": true,
  "sections": [
    {
      "className": "CSE_1_A",
      "branch": "CSE",
      "year": 1,
      "section": "A",
      "enrolledCount": 45,
      "faculty": {
        "name": "Prof. Ananya Rao",
        "email": "cse.teacher1@college.edu"
      },
      "todayStatus": "SUBMITTED", // "SUBMITTED" or "PENDING"
      "submittedAt": "2026-09-19T09:45:00.000Z",
      "presentCount": 41,
      "absentCount": 4,
      "attendanceRate": 91.1,
      "monthlyAverage": 87.8
    }
  ]
}
```

```
}
```

3.5 Section Details & Student Attendance Matrix

GET /api/dashboard/section-details/CSE_1_A?date=2026-09-19

Fetches the full class register matrix, day entries, and today's marked absentees with parent phone numbers.

```
// Response (200 OK)
{
  "success": true,
  "section": {
    "className": "CSE_1_A",
    "faculty": { "name": "Prof. Ananya Rao", "email": "cse.teacher1@college.edu" },
    "date": "2026-09-19",
    "totalStudents": 45,
    "todayPresent": 41,
    "todayAbsent": 4,
    "absentees": [
      { "rollNo": "103", "name": "Rohan Verma", "parentPhone": "+919876543213" },
      { "rollNo": "107", "name": "Neha Sharma", "parentPhone": "+919876543217" }
    ],
    "studentsMatrix": [
      {
        "studentId": "66da9123f8b1c4e901a1bc40",
        "rollNo": "101",
        "name": "Aarav Sharma",
        "parentPhone": "+919876543210",
        "totalHeld": 22,
        "presentCount": 20,
        "absentCount": 2,
        "percentage": 90.9,
        "todayStatus": "P" // "P" or "A"
      }
    ]
  }
}
```

3.6 Student Directory with Real-time Percentage

GET /api/dashboard/students?department=CSE&status;=below75&search;=Aarav&page;=1&limit;=20

Filterable and searchable student list across the institution with computed attendance standing.

```
// Response (200 OK)
{
  "success": true,
  "total": 1,
  "page": 1,
  "limit": 20,
  "students": [
    {
      "id": "66da9123f8b1c4e901a1bc40",
      "rollNo": "101",
      "name": "Aarav Sharma",
      "className": "CSE_1_A",
      "branch": "CSE",
      "year": 1,
      "section": "A",
      "parentPhone": "+919876543210",
      "totalConducted": 24,
      "totalAttended": 22,
      "percentage": 91.7,
      "statusCategory": "GOOD" // "GOOD" (>=75%) | "SHORTAGE" (65-74%) | "CRITICAL" (<65%)
    }
  ]
}
```

3.7 Individual Student 360° Profile

GET /api/dashboard/student/:id

Complete history for student counseling, parent meetings, and attendance shortage reviews.

```
// Response (200 OK)
{
  "success": true,
  "student": {
    "id": "66da9123f8b1c4e901a1bc40",
    "name": "Aarav Sharma",
    "rollNo": "101",
    "className": "CSE_1_A",
    "parentPhone": "+919876543210",
    "overallStats": {
      "totalClasses": 45,
      "classesAttended": 40,
      "classesAbsent": 5,
      "percentage": 88.9,
      "status": "GOOD"
    },
    "monthlyBreakdown": [
      { "month": "August", "year": 2026, "total": 22, "attended": 20, "percent": 90.9 },
      { "month": "September", "year": 2026, "total": 23, "attended": 20, "percent": 87.0 }
    ],
    "recentAbsences": [
      { "date": "2026-09-15", "recordedBy": "Prof. Ananya Rao", "smsStatus": "sent" }
    ]
  }
}
```

3.8 Low Attendance & Defaulters Alert List

GET /api/dashboard/defaulters?threshold=75&department;=ALL

Lists all students below mandatory attendance threshold with calculated shortage days needed to recover.

```
// Response (200 OK)
{
  "success": true,
  "threshold": 75,
  "totalDefaulters": 18,
  "defaulters": [
    {
      "studentId": "66da9123f8b1c4e901a1bc43",
      "rollNo": "104",
      "name": "Vikram Singh",
      "className": "CSE_1_A",
      "branch": "CSE",
      "parentPhone": "+919876543214",
      "totalHeld": 30,
      "attended": 18,
      "percentage": 60.0,
      "daysNeededFor75": 18,
      "riskLevel": "CRITICAL"
    }
  ]
}
```

3.9 Faculty Submission Compliance Tracker

GET /api/dashboard/faculty-status?date=2026-09-19&department;=ALL

Empowers Dean and HODs to verify punctuality of attendance submission across all periods and classes.

```
// Response (200 OK)
{
  "success": true,
  "date": "2026-09-19",
  "summary": { "totalClasses": 12, "submitted": 10, "pending": 2 },
  "facultyList": [
    {
      "className": "CSE_1_A",
      "facultyName": "Prof. Ananya Rao",
      "facultyEmail": "cse.teacher1@college.edu",
      "status": "SUBMITTED",
      "submissionTime": "09:42 AM",
      "presentRate": 91.1
    },
    {
      "className": "ECE_2_B",
      "facultyName": "Prof. Rajesh Kumar",
      "facultyEmail": "ece.teacher2@college.edu",
      "status": "PENDING",
      "submissionTime": null,
      "presentRate": null
    }
  ]
}
```

3.10 Time-Series Analytics (Charts & Graphs)

GET /api/dashboard/trends?days=14&department;=ALL

Returns structured data ready to pass directly into **Recharts** <LineChart> or <BarChart>.

```
// Response (200 OK)
{
  "success": true,
  "days": 14,
  "trend": [
    { "date": "2026-09-08", "day": "Tue", "attendanceRate": 89.2, "present": 375, "absent": 45 },
    { "date": "2026-09-09", "day": "Wed", "attendanceRate": 91.4, "present": 384, "absent": 36 },
    { "date": "2026-09-10", "day": "Thu", "attendanceRate": 87.0, "present": 365, "absent": 55 }
  ],
  "dayOfWeekDistribution": {
    "Mon": 86.4, "Tue": 90.1, "Wed": 91.2, "Thu": 88.5, "Fri": 84.1, "Sat": 82.0
  }
}
```

4. Frontend Integration Guide for React Developer

To build a responsive, modern executive portal for Dean Ma'am and HODs, follow these best practices:

Aspect	Recommended Practice & Library
Data Fetching	Use Axios with a configured Axios instance (baseUrl: http://172.29.58.78:3000) and request interceptor attaching Authorization: Bearer \${localStorage.getItem('token')}.
Charts & Graphs	Use Recharts (<ResponsiveContainer>, <LineChart>, <BarChart>). Data from /api/dashboard/trends feeds directly into the data prop without manual transformation.
UI Components	Tailwind CSS / Lucide React: Create 4 core page views: 1. Executive Overview (Cards, Charts, Department grid) 2. Section Matrix (Class selector, student matrix table) 3. Defaulters Console (Table with badge tags <75% and quick parent call links) 4. Faculty Compliance (Live checklist of who has and hasn't submitted today).
Role Routing	If user.role === 'hod', pre-select and lock the Department dropdown to their assigned department (user.department). If 'dean' or 'principal', enable the full college dropdown filter.

5. Standard Error Responses

```
// 401 Unauthorized (Missing or Invalid Token)
{ "success": false, "message": "Authentication token is required" }

// 403 Forbidden (Role mismatch)
{ "success": false, "message": "You do not have permission for this action" }

// 404 Not Found
{ "success": false, "message": "Student or section not found" }
```